THE EXACT SEQUENCE COMPARING THE FIRST COHOMOLOGY OF A MODIFICATION WITH ITS BLOWUP

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file resolve.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the computation of the direct images and the identification of the support of the first one, the global generation statement, and the degeneration of the Leray spectral sequence.

Editorial note. The label lemma-exact-sequence names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source resolve.tex –label lemma-exact-sequence –start 1900 –end 1944 –local-id 28102 returned RESULT=OK (L28102 free, occupancy FREE, public interval clear). Pre-assembly tag self-check: canonical tag resolved and not present in reference/public-items.jsonl.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Structure sheaf of the modification and vanishing of higher direct images. More on Morphisms, the lemma that a dominant birational morphism to a normal scheme has structure sheaf pushforward the structure sheaf, and Cohomology of Schemes, the lemma that higher direct images vanish above the dimension of the fibres. Identify the zeroth pushforward and kill the higher ones, leaving only the first direct image in the Leray spectral sequence.
- Global generation of a coherent sheaf with zero dimensional support. Cohomology of Schemes, the lemma on coherent sheaves supported in dimension zero. Shows that the first direct image is generated by global sections.
- Vanishing of higher cohomology of coherent sheaves on a modification. The observations following the vanishing situation, giving vanishing of cohomology in degrees larger than one for coherent sheaves on the source. Collapses the Leray spectral sequence to the short exact sequence of the statement.

Source reference. Original lemma 0B3J: sources/stacks-chow/tags.txt, source lines 13967–13967. Prerequisite pointer invoked at proof line(s) 1938; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:resolve.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

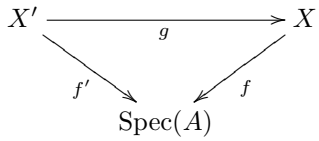
Source reference. Original lemma 02V7: sources/stacks-chow/tags.txt, source lines 3758–3758. Prerequisite pointer invoked at proof line(s) 1929; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 01F2: sources/stacks-chow/tags.txt, source lines 1855–1855. Prerequisite pointer invoked at proof line(s) 1943; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AY8: sources/stacks-chow/tags.txt, source lines 13781–13781. Prerequisite pointer invoked at proof line(s) 1925; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 07UY: sources/stacks-resolve/pinned-original-source.tex, source lines 1584–1584. Prerequisite pointer invoked at proof line(s) 1910; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $(A, \mathfrak{m}, \kappa)$ be a Noetherian normal local domain of dimension 2. Consider a commutative diagram*



where f and f' are modifications as in Situation 07UY and X normal. Then we have a short exact sequence

$$0 \to H^1(X, \mathcal{O}_X) \to H^1(X', \mathcal{O}_{X'}) \to H^0(X, R^1 g_* \mathcal{O}_{X'}) \to 0$$

Also $\dim(\mathit{Supp}(R^1 g_* \mathcal{O}_{X'})) = 0$ *and $R^1 g_* \mathcal{O}_{X'}$ is generated by global sections.*

Proof. We will use the observations made following Situation 07UY without further mention. As X is normal and g is dominant and birational, we have $g_* \mathcal{O}_{X'} = \mathcal{O}_X$, see for example More on Morphisms, Lemma 0AY8. Since the fibres of g have dimension ≤ 1, we have $R^p g_* \mathcal{O}_{X'} = 0$ for $p > 1$, see for example Cohomology of Schemes, Lemma 02V7. The support of $R^1 g_* \mathcal{O}_{X'}$ is contained in the set of points of X where the fibres of g' have dimension ≥ 1. Thus it is contained in the set of images of those irreducible components $C' \subset X'_s$ which map to points of X_s which is a finite set of closed points (recall that $X'_s \to X_s$ is a morphism of proper 1-dimensional schemes over κ). Then $R^1 g_* \mathcal{O}_{X'}$ is globally generated by Cohomology of Schemes, Lemma 0B3J. Using the morphism $f : X \to S$ and the references above we find that $H^p(X, \mathcal{F}) = 0$ for $p > 1$ for any coherent $\mathcal{O}_X$-module $\mathcal{F}$. Hence the short exact sequence of the lemma is a consequence of the Leray spectral sequence for g and $\mathcal{O}_{X'}$, see Cohomology, Lemma 01F2. $\square$